DIPLOMA IN ELECTRONICS / ELECTRONICS AND COMMUNICATION ENGINEERING • SEMESTER 3

Principles of Electronic Communication

This corrected edition treats 3042 as an engineering communication textbook. It explains AM, FM, sampling, pulse modulation, antennas, transmitters, receivers, noise and demodulation with block diagrams, spectra, formulas, numerical problems and answer patterns.

COURSE CODE 3042	CREDITS 3
PERIODS 45	TYPE Program Core

□□□ communication subject □□□□ formula list □□□□□□ □□□□. Signal, carrier, spectrum, bandwidth, power, noise, antenna, transmitter, receiver flow □□□□□□ diagram □□□□□ □□□□□□□□.

COURSE BLUEPRINT

What this handbook is expected to teach

This section converts the dull syllabus table into a usable engineering route: prerequisite knowledge, outcome target, field meaning, diagrams to draw, and problems to solve.

Official objectives

- ✓ Introduce electronic communication and need of modulation.
- ✓ Explain AM, FM and pulse modulation schemes.
- ✓ Understand antenna role and radiation concepts.
- ✓ Explain working of AM and FM transmitters and receivers.

Prerequisite stack

- ✓ Active and passive component basics.
- ✓ Power, RMS, peak values and signal representation.
- ✓ Integration, differentiation, trigonometry and mathematical signal forms.
- ✓ Amplifiers, oscillators and switching circuits.

How to use

1. Read the concept explanation.
2. Study the diagram or waveform.
3. Solve the worked example without looking.
4. Write the expected-answer format.

CO	Outcome	Hours	Engineering meaning
CO 1	Solve AM and FM parameters from signal data.	11	Calculate modulation index, sideband frequency, bandwidth and transmitted power.
CO 2	Explain pulse modulation and antennas.	11	Understand sampling, PAM, PWM, PPM, PCM and antenna parameters.
CO 3	Explain AM and FM transmitters.	11	Draw blocks of transmitters, modulators, pre-emphasis and noise sources.
CO 4	Illustrate AM and FM receivers.	10	Explain superheterodyne receiver, IF, detector, AGC and receiver characteristics.

REDESIGNED SYLLABUS STRUCTURE

Module map with diagrams, applications and problem targets

The old plain syllabus table is not enough. This map tells what to draw, what to explain, where the marks come from, and how the concept is used in industry.

Module	Core topics	Must draw / visualize	Must solve / apply	Hours
M1: Amplitude modulation and frequency modulation parameters	Modulation concept, AM wave, AM power, sidebands, DSBSC, SSB, VSB, FM spectrum and comparison.	AM waveform, AM spectrum, FM spectrum, modulation classification tree.	AM index, sideband frequencies, AM power, bandwidth, FM deviation and Carson bandwidth.	11
M2: Sampling, pulse modulation and antenna fundamentals	Sampling theorem, PAM, PWM, PPM, PCM, radiation concept, antenna parameters, microwave and microstrip antennas.	Sampling waveform, PAM/PWM/PPM comparison, antenna radiation pattern.	Minimum sampling rate and Nyquist rate problems.	11
M3: AM and FM transmitters, modulators and noise	AM low level and high level transmitters, collector modulator, balanced modulator, Armstrong FM transmitter, pre-emphasis, de-emphasis and noise.	AM transmitter block, FM transmitter block, collector modulator, balanced modulator, pre-emphasis circuit.	SNR and noise comparison concept questions.	11
M4: AM and FM receivers, demodulation and AGC	Receiver characteristics, superheterodyne receiver, AM and FM receiver blocks, choice of IF, diode detector, simple and delayed AGC.	Superheterodyne receiver, AM receiver block, FM receiver block, diode detector and AGC block.	Receiver characteristics and IF reasoning questions.	10

Amplitude modulation and frequency modulation parameters

Modulation is used because baseband signals cannot be transmitted efficiently over long distance. The message is shifted to a high frequency carrier.

Textbook explanation

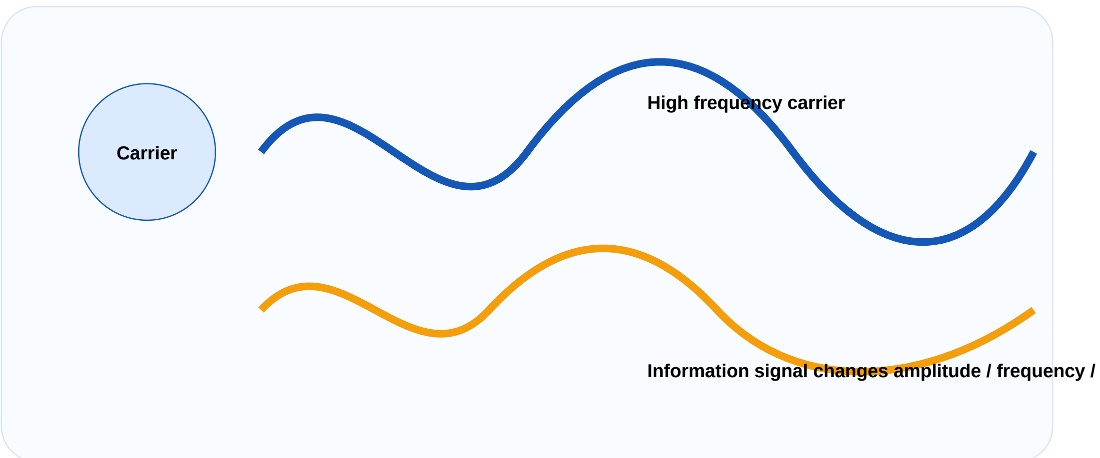
In amplitude modulation, carrier amplitude changes according to the message while frequency and phase remain constant.

AM spectrum contains carrier, upper sideband and lower sideband. The sidebands carry information. Carrier consumes power but does not carry information.

In frequency modulation, carrier frequency changes according to the message amplitude. FM has better noise immunity than AM.

Bandwidth, modulation index and power calculations are the scoring area of this module.

> AM/FM □□□□□□□□□□□□□ waveform, spectrum, formula, unit □□□□□□□ □□□□□□□□□□□□ □□□□□□□□□. Sideband frequency □□□□□□□□□□□□ numerical □□□□□□□□□ □□□□□□□□.



Carrier plus information signal and modulated signal idea.

Engineering subtopics

Need of modulation

Makes antenna size practical and avoids mixing of signals.

- ✓ Frequency translation.
- ✓ Multiplexing possible.
- ✓ Radiation efficiency improves.

AM wave

Amplitude of carrier varies with message.

- ✓ Modulation index m.
- ✓ Carrier plus two sidebands.
- ✓ $BW = 2f_m$.

AM power

Power is distributed in carrier and sidebands.

- ✓ Carrier power is constant.
- ✓ Sideband power increases with m.
- ✓ Over modulation causes distortion.

FM wave

Frequency deviation represents message.

- ✓ Modulation index $\beta = \Delta f/f_m$.
- ✓ Better noise performance.
- ✓ Uses limiter in receiver.

DSBSC, SSB, VSB

Bandwidth and power saving forms.

- ✓ DSBSC suppresses carrier.
- ✓ SSB transmits one sideband.
- ✓ VSB partly transmits one sideband.

AM vs FM

Compare bandwidth, noise and circuit complexity.

- ✓ AM narrow bandwidth.
- ✓ FM wider bandwidth.
- ✓ FM high fidelity.

Important formulas / rules

AM modulation index: $m = V_m/V_c$

AM bandwidth: $BW = 2f_m$

AM total power: $P_t = P_c(1 + m^2/2)$

FM modulation index: $\beta = \Delta f/f_m$

Carson rule: $BW = 2(\Delta f + f_m)$

Common mistakes

- ✓ Confusing carrier frequency and modulating frequency.
- ✓ Forgetting that AM has two sidebands.
- ✓ Using percentage modulation without converting to decimal.
- ✓ Writing FM bandwidth as 2fm for every case.

Worked examples inside this module

Carrier is 1 MHz and audio is 5 kHz. Find AM sidebands.

USB = 1000 kHz + 5 kHz = 1005 kHz. LSB = 1000 kHz - 5 kHz = 995 kHz.

FM deviation is 75 kHz and modulating frequency is 15 kHz. Find bandwidth using Carson rule.

BW = 2(75 + 15) kHz = 180 kHz.

Sampling, pulse modulation and antenna fundamentals

Pulse modulation prepares analog information for digital and time-sampled communication. Antenna is the conversion point between circuit signal and electromagnetic wave.

Textbook explanation

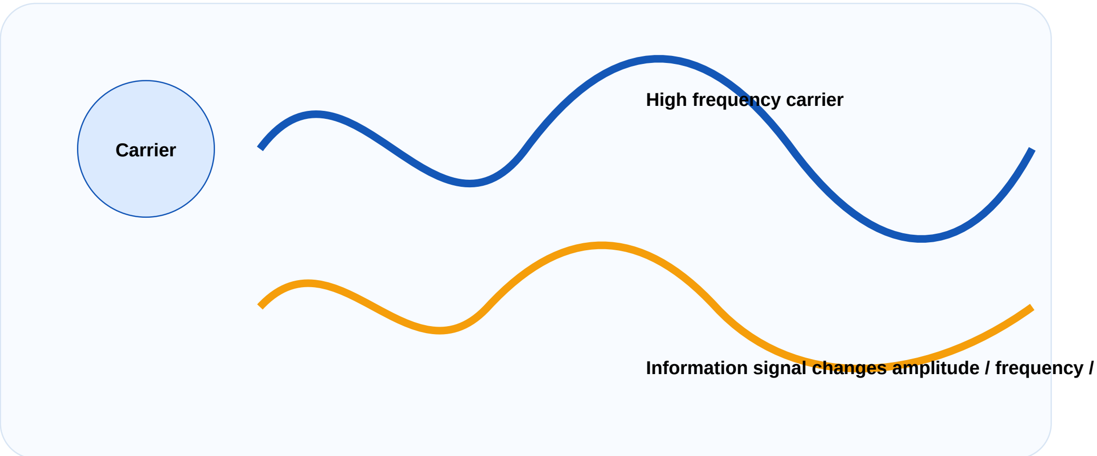
Sampling theorem states that sampling frequency must be at least twice the highest frequency present in the message.

PAM changes pulse amplitude, PWM changes pulse width, PPM changes pulse position and PCM represents samples using digital code.

Antenna radiation pattern shows how energy is radiated in space. Antenna parameters include gain, directivity, bandwidth, polarization and radiation resistance.

Microstrip antennas are compact and widely used in modern communication devices.

> Sampling theorem simple □□□□□□□□□□□ practical importance □□□□□□□□. $f_s < 2f_m$ □□□□□□□□□ aliasing □□□□.



Sampling and pulse modulation as time-domain representation.

Engineering subtopics

Sampling theorem

Minimum sampling rate prevents aliasing.

- ✓ $f_s \geq 2f_m$.
- ✓ Nyquist rate is 2fm.
- ✓ Anti-aliasing filter is used before sampling.

PAM

Pulse amplitude varies with signal.

- ✓ Simple but noise sensitive.
- ✓ Used as intermediate step in PCM.

PWM

Pulse width varies with signal.

- ✓ Amplitude remains constant.
- ✓ Useful in power and control systems.

PPM

Pulse position varies with signal.

- ✓ Good noise immunity.
- ✓ Needs synchronization.

PCM

Sample, quantize and encode.

- ✓ Digital transmission.
- ✓ Needs higher bandwidth.
- ✓ Good noise immunity.

Antenna parameters

Gain, radiation pattern, directivity and polarization.

- ✓ Antenna length depends on wavelength.
- ✓ Radiation pattern shows direction.
- ✓ Microstrip antenna is low-profile.

Important formulas / rules

Sampling condition: $f_s \geq 2f_m$

Wavelength: $\lambda = c/f$

Approximate half-wave antenna length: $L = \lambda/2$

Common mistakes

- ✓ Writing sampling frequency equal to signal frequency.
- ✓ Not explaining aliasing.
- ✓ Confusing PAM and PWM.
- ✓ Drawing antenna as only a wire without radiation pattern.

Worked examples inside this module

An audio signal has highest frequency 4 kHz. Find minimum sampling frequency.

$f_s = 2f_m = 8$ kHz.

Find wavelength for 100 MHz signal.

$\lambda = c/f = 3\times10^8 / 100\times10^6 = 3$ m.

AM and FM transmitters, modulators and noise

A transmitter is not one amplifier. It is an arranged chain of oscillator, modulator, frequency multiplier, power amplifier and antenna matching blocks.

Textbook explanation

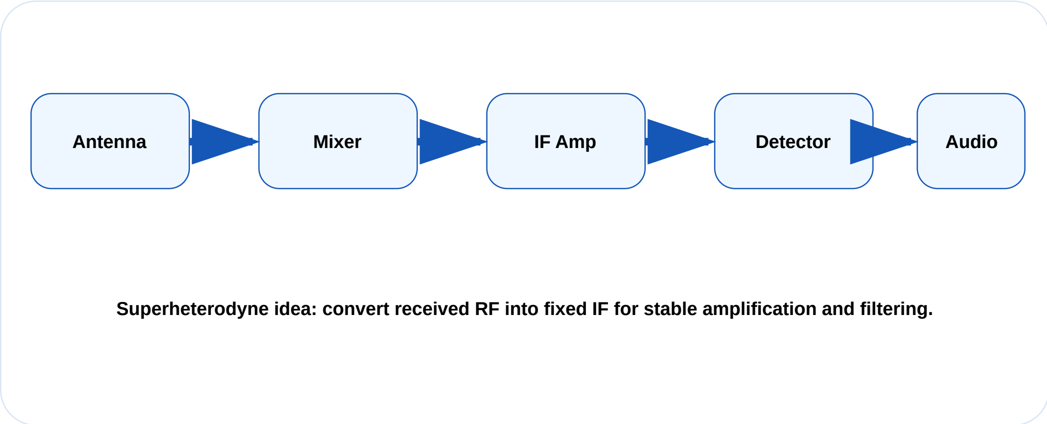
Low-level AM modulation is done before power amplification. High-level AM modulation is done at high power stage.

Balanced modulator suppresses carrier and generates DSBSC signal. Collector modulator varies collector voltage of RF power amplifier.

Armstrong method generates FM indirectly using phase modulation and frequency multiplication.

Noise may be external or internal. FM uses pre-emphasis and de-emphasis to improve high-frequency SNR.

> Transmitter block diagram answer-□ signal flow arrow, block purpose, modulation point, antenna output □□□□□□□ □□□□□□□.



Transmitter and receiver block thinking.

Engineering subtopics

AM transmitter

Generates and amplifies AM signal.

- ✓ Low-level modulation.
- ✓ High-level modulation.
- ✓ RF power amplifier feeds antenna.

Collector modulator

Modulates collector supply of RF amplifier.

- ✓ High level AM.
- ✓ Good power handling.

Balanced modulator

Suppresses carrier.

- ✓ Used for DSBSC.
- ✓ Saves carrier power.

FM transmitter

Armstrong method and frequency multiplication.

- ✓ Stable crystal oscillator.
- ✓ Phase modulation to FM conversion.
- ✓ Frequency multipliers increase deviation.

Pre-emphasis

Boosts high audio frequencies before transmission.

- ✓ Improves SNR.
- ✓ Receiver uses de-emphasis.

Noise

Unwanted signal in communication system.

- ✓ External noise.
- ✓ Internal noise.
- ✓ Signal-to-noise ratio.

Important formulas / rules

SNR = signal power / noise power

SNR in dB = $10 \log_{10}(S/N)$

Noise figure indicates degradation of SNR by a receiver or stage.

Common mistakes

- ✓ Mixing modulator and demodulator.
- ✓ Not differentiating low-level and high-level AM transmitter.
- ✓ Forgetting de-emphasis in receiver.
- ✓ Writing only noise definition without types.

Worked examples inside this module

Why balanced modulator is used?

It suppresses carrier and produces DSBSC signal, saving transmitted power compared with ordinary AM.

Why pre-emphasis is used in FM?

Noise effect is higher at high audio frequencies, so high-frequency audio components are boosted before transmission and corrected by de-emphasis at receiver.

AM and FM receivers, demodulation and AGC

Receiver quality is measured using selectivity, sensitivity, fidelity and noise figure. The superheterodyne receiver converts any selected RF station to a fixed IF.

Textbook explanation

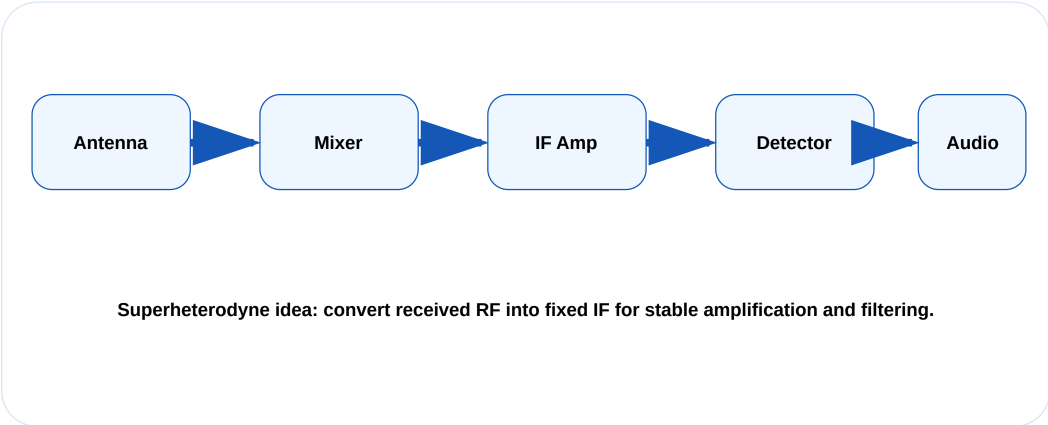
Selectivity is ability to reject unwanted nearby signals. Sensitivity is ability to receive weak signals. Fidelity is faithful reproduction of original signal.

Superheterodyne receiver uses mixer and local oscillator to produce fixed IF. Fixed IF gives stable gain and filtering.

AM demodulation can be done using diode detector. AGC keeps output nearly constant when received signal strength changes.

FM receiver includes limiter, discriminator or detector and de-emphasis network.

> Receiver answer-□ RF amplifier, mixer, local oscillator, IF amplifier, detector, AF amplifier □□□□ flow □□□□.



Superheterodyne receiver chain.

Engineering subtopics

Selectivity

Ability to select desired signal.

✓ Depends on tuned circuits and IF filters.

Sensitivity

Ability to receive weak signal.

✓ Improved by RF and IF gain.

Fidelity

Faithful reproduction of original signal.

✓ Depends on bandwidth and distortion.

Superheterodyne

RF converted to fixed IF.

✓ Mixer plus local oscillator.

✓ IF amplification gives stable performance.

AM detector

Diode detector recovers envelope.

✓ Needs RC time constant selection.

✓ Used for AM demodulation.

AGC

Automatic gain control.

- ✓ Simple AGC.
- ✓ Delayed AGC for stronger signals.

Important formulas / rules

For high-side injection: $f_{LO} = f_{RF} + f_{IF}$

For low-side injection: $f_{LO} = f_{RF} - f_{IF}$

Receiver quality terms: selectivity, sensitivity, fidelity and noise figure.

Common mistakes

- ✓ Spelling and concept error: superheterodyne, not superhetrodyne.
- ✓ Not explaining why IF is used.
- ✓ Drawing transmitter block when asked receiver.
- ✓ Confusing AGC with audio volume control.

Worked examples inside this module

Why fixed IF is used in superheterodyne receiver?

Fixed IF allows high gain and sharp filtering at one constant frequency for all received stations.

What is the function of AGC?

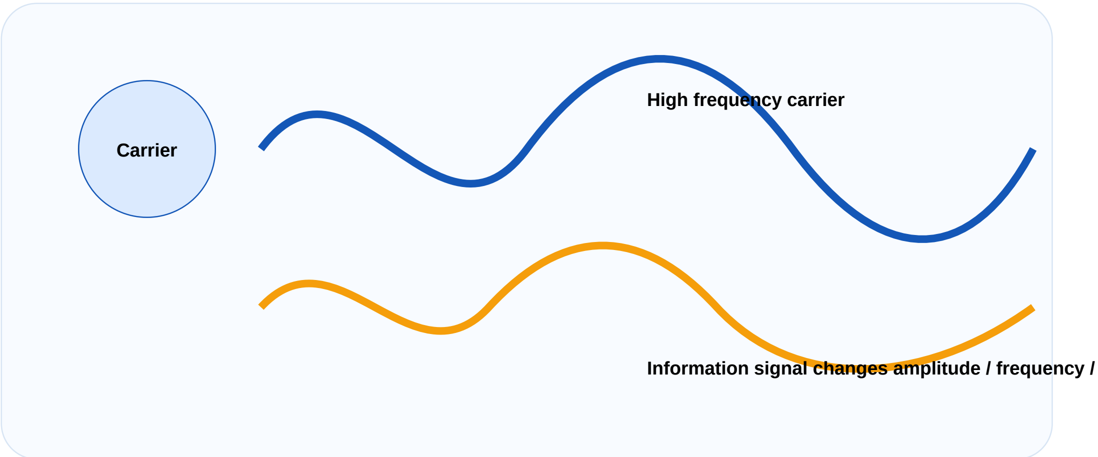
AGC changes receiver gain automatically so output remains nearly constant even when input signal strength varies.

ILLUSTRATION LAB

Images, block diagrams and waveform thinking

For technical subjects, the diagram is half of the answer. These visuals make the lesson look and work like a textbook instead of a plain note.

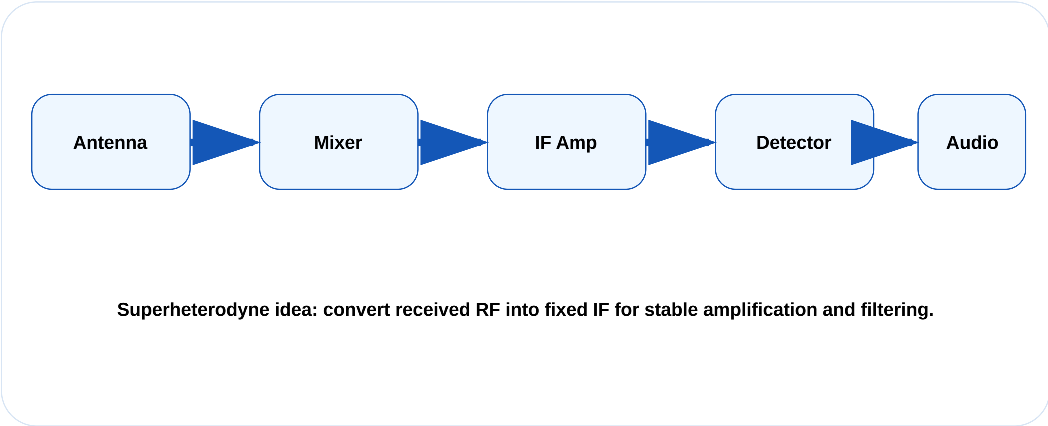
Modulation concept



Carrier is modified by information signal.

This diagram supports AM, FM and PM explanation.

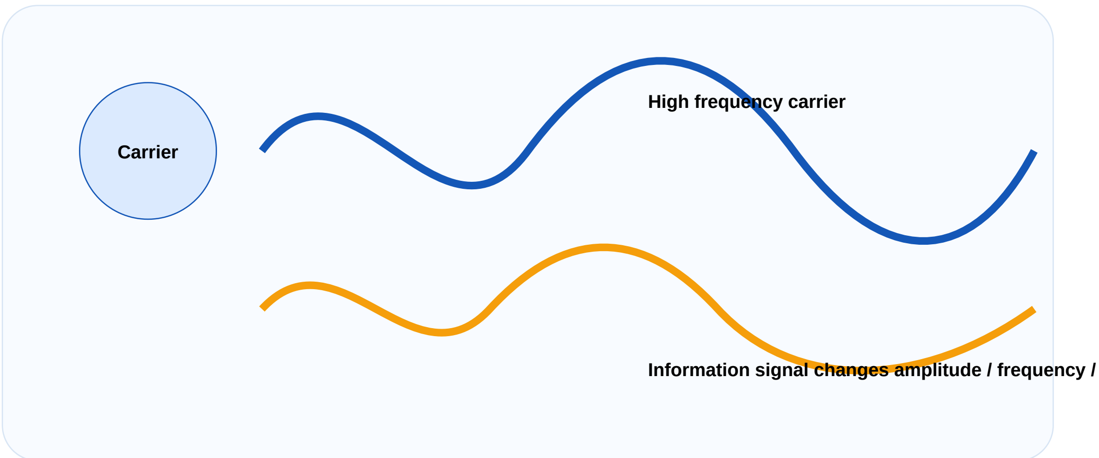
Superheterodyne receiver



RF is converted into fixed IF before detection.

Use this block in receiver answers.

Spectrum thinking



Sidebands carry information around carrier.

Most AM numerical questions come from this view.

General signal chain

Source, transmitter, channel, receiver and destination

Diagram, waveform, block flow and practical meaning should be studied together.

Source, transmitter, channel, receiver and destination.

Never write transmitter or receiver answers without signal flow.

PROBLEM LAB

Solved numerical and design problems

Each problem shows the method, not only the answer. This is the difference between a notebook and an engineering handbook.

AM sidebands

Problem: Carrier frequency is 800 kHz and modulating frequency is 5 kHz. Find sideband frequencies.

1. Upper sideband = fc + fm.
2. Lower sideband = fc - fm.
3. Substitute fc = 800 kHz and fm = 5 kHz.

Answer: USB = 805 kHz, LSB = 795 kHz

AM total power

Problem: Carrier power is 100 W and modulation index is 0.6. Find total AM power.

1. Use $P_t = P_c(1 + m^2/2)$.
2. $m^2/2 = 0.36/2 = 0.18$.
3. $P_t = 100 \times 1.18$.

Answer: 118 W

FM bandwidth

Problem: Frequency deviation is 60 kHz and maximum audio frequency is 10 kHz. Find bandwidth.

1. Use Carson rule $BW = 2(\Delta f + f_m)$.
2. Substitute 60 kHz and 10 kHz.

Answer: 140 kHz

Sampling rate

Problem: Highest frequency in message is 3.4 kHz. Find minimum sampling frequency.

1. Use Nyquist sampling condition.
2. f_s must be at least $2f_m$.

Answer: 6.8 kHz minimum

PRACTICE

Expected questions by module

Module 1

- Classify modulation techniques.
- Derive AM wave expression and draw AM spectrum.
- Solve AM modulation index, bandwidth and power problems.
- Explain DSBSC, SSB and VSB.
- Explain FM spectrum and Carson bandwidth.
- Compare AM and FM.

Module 2

- State sampling theorem and explain its significance.
- Compare PAM, PWM, PPM and PCM.
- Explain aliasing and Nyquist rate.
- Explain radiation pattern and antenna parameters.
- Explain microwave and microstrip antennas.

Module 3

- Explain AM transmitter low-level and high-level types.
- Explain collector modulator.
- Explain balanced modulator.
- Explain Armstrong FM transmitter.
- Explain pre-emphasis and de-emphasis.
- Classify noise in communication systems.

Module 4

- Define selectivity, sensitivity, fidelity and noise figure.
- Explain superheterodyne receiver with block diagram.
- Explain AM receiver and FM receiver blocks.
- Explain choice of IF in receivers.
- Explain diode detector and AGC.
- Compare AM and FM receivers.

MODEL PAPER

Full practice question paper and answer guide

This model follows the syllabus order and avoids copying official papers.

COURSE 3042 PRINCIPLES OF ELECTRONIC COMMUNICATION
Time: 3 Hours   Maximum Marks: 100

PART A
1. Define modulation.

2. What is modulation index in AM?
3. Define frequency deviation.
4. State sampling theorem.
5. Name four pulse modulation schemes.
6. Define antenna gain.
7. What is pre-emphasis?
8. Define signal-to-noise ratio.
9. What is selectivity?
10. What is AGC?

PART B

11. Explain need of modulation.
12. Derive AM bandwidth and draw spectrum.
13. Solve AM power problem from given carrier power and m.
14. Compare PAM, PWM, PPM and PCM.
15. Explain antenna parameters.
16. Explain AM transmitter block diagram.
17. Explain balanced modulator.
18. Explain receiver characteristics.

PART C

19. Explain AM and FM techniques with formulas and comparison.
20. Explain sampling theorem, aliasing and PCM.
21. Explain microwave and microstrip antennas.
22. Explain Armstrong FM transmitter and noise reduction.
23. Explain superheterodyne receiver and choice of IF.
24. Explain diode detector and AGC.
25. Compare AM and FM receivers.
26. Solve one AM numerical and one FM bandwidth numerical.

Answer writing guide

- ✓ For modulation questions draw time waveform and frequency spectrum.
- ✓ For numerical answers write formula, substitution, unit and final answer.
- ✓ For antenna questions include radiation pattern and parameters.
- ✓ For receiver questions draw block diagram and explain each block in order.

REFERENCES

Books and resources

Type	Reference
T1	Electronic Communication Systems - George Kennedy - TMH
T2	Electronic Communications - Roddy and Coolen - PHI
R1	Electronic Communication Systems - Frank R Dungan
R2	Electronic Communication Systems - Wayne Tomasi
R3	Communication Systems Analog and Digital - Sanjay Sharma
Online	NPTEL 117105143 and selected communication system resources

Final revision rule: Prepare one diagram, one formula, one application and one common mistake for every major topic.